Summer 2027 internship availability: May 17–August 6, 2027 (12 weeks, full-time)

Lei Zhao

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ML Systems | **Distributed Training** | **GNN** | **RecSys** | **CXL Memory Pool** | **RISC-V** | **Linux kernel** | **AI Agent**

EDUCATION

Texas State University *(Expected graduation: May 2028)* **08/24 – Present**
Ph.D. Candidate in Computer Science (Advisor: Dr. Chul-Ho Lee)

China University of Geosciences – M.Eng. in Electronic and Information **09/21 – 06/24**

WORK EXPERIENCE

SK hynix America – *Advanced Memory System Research Intern* **05/26 – 08/26**

- **Developed Linux kernel patch** to expose CXL memory as Device DAX on an AMD server without Special Purpose Memory (SPM) support.
- **Developed CXL shared-memory library** with C/C++ APIs and PyTorch bindings, implementing software-managed cache coherence.
- **Offloaded graph and feature data to CXL memory pool**, reducing GNN training time.
- **Investigated CXL+RDMA multipath data transfer** to mitigate workload skew.
- **Explored CXL memory pooling** for PyTorch DDP gradient sync using a custom hook.
- **Implemented and evaluated experimental CXL-backed NCCL collectives**.
- **Benchmarked NVSHMEM against CXL shared memory**.
- **Evaluated NVSHMEM's UCX transport** for cross-host GPU communication, including GDRCopy integration and memory-registration diagnostics.
- **Explored CXL-backed KV cache offloading** for SGLang prefill–decode disaggregation.
- **Investigated read amplification caused by sparse graph data access**.
- **Set up an on-premises 480B Qwen coding agent service** using vLLM with tensor parallelism.
- **Developed team tools and resolved server issues**, addressing APT update failures, VSCodium remote connection issues, PSU and BMC problems. [VS code extension]

SELECTED PROJECTS

FROST: A Unified Multi-GPU Training Framework for Temporal Graph Neural Networks [Paper] [Code] (*Accepted at SIGMOD 2027*, CCF A, CORE A*) **08/25 – 04/26**

- Developed a **multi-GPU training pipeline** using multiple threads, processes, and CUDA streams for 8 memory-based/memoryless TGNN models, achieving up to **12.61× speedup over TGL**.
- **Designed Planner-Syncer mechanism** to analyze cross-batch node-state dependencies and route updated states through same-GPU reuse, GPUDirect transfer, or async host-memory write-back.
- **Identified an idle inter-GPU communication window** between node-state updates and the first gradient sync, and implemented a custom hook to overlap state transfers with remaining forward and early backward computation.
- **Developed shared-memory ring buffers** for zero-copy host-to-worker graph data access, scattering only lightweight metadata to avoid bulk graph serialization.
- **Implemented topology-aware NUMA placement**, binding GPU workers and per-GPU shared-memory buffers to GPU-local NUMA nodes before first touch.

RISCuda: GPU-Resident LLM Tool Execution [Code] **Ongoing**

- Enabled **on-GPU tool calling without CPU invocation or dispatch per tool call**: Qwen generation, tool-call detection, C calculator execution via a CUDA RISC-V interpreter, and result feedback run on the GPU within a **single host CUDA Graph launch**.
- C-calculator tests achieved **1.71× speedup over Qwenputer**.

Accelerating cuGraph GNN Training via CXL Memory Pool Data Offloading **05/26 – Present**
Collaborative Project with NVIDIA cuGraph team during internship

- **Profiled cuGraph training** to measure per-stage execution time, identify performance bottlenecks, and count sync barriers.
- **Implemented CXLGraph, a data loader, and a neighbor sampler** to offload graph data to a CXL memory pool and eliminate RDMA sync barriers in cuGraph's distributed graph sampling.
- **Offloaded feature data to a CXL memory pool** by replacing NVIDIA's distributed contiguous memory backend, eliminating RDMA sync barriers in cuGraph's remote feature gathering.
- **Implemented versioned tensors** to maintain consistency during mutable embedding updates.
- Developed and ran CI tests; contributed to cuDF/cuGraph (cuDF PR; cuGraph PR).

Building Ultra-Low-Cost GPU Computing Servers **11/22 – 04/23**

- Built multiple GPU servers, including one with **15 NVIDIA P102-100 mining GPUs**.
- Configured Xubuntu, networking, and NFS; automated deployment with Dockerfiles.
- Profiled and tuned performance with monitoring tools, bcache, and swap.

SELECTED PUBLICATIONS & MANUSCRIPTS

marks descriptive titles used for manuscripts under review.

- **FROST: A Unified Multi-GPU Training Framework for Temporal Graph Neural Networks** [Code], **Lei Zhao**, Joseph Nke, Jongryool Kim, and Chul-Ho Lee (*Accepted at SIGMOD 2027*, CCF A, CORE A*)
- **Accelerating SpGEMM with HBM Logic Die Processing** (*Second author, Under review*)
- **A CUDA Graph Analytics Primitive** (*Second author, Under review*)
- **A CUDA Graph Data Structure** (*Second author, Under review*)

SKILLS

- **Programming:** Python, C/C++, CUDA, OpenMP, Shell.
- **MLSys:** PyTorch DDP, NCCL, CUDA streams, pipeline optimization, PyTorch/Nsight profiling.
- **Systems:** Linux, CXL/Device DAX, NUMA, shared memory, Docker, Git.